

Modeling of ATP–ADP steady-state exchange rate mediated by the adenine nucleotide translocase in isolated mitochondria

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Keywords

adenine nucleotide carrier; adenine nucleotide translocator; ATP synthasome; ATP/ADP carrier; systems biology

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(Received 3 June 2009, revised 20 September 2009, accepted 23 September 2009)

doi:10.1111/j.1742-4658.2009.07394.x

A computational model for the ATP–ADP steady-state exchange rate mediated by adenine nucleotide translocase (ANT) versus mitochondrial membrane potential dependence in isolated rat liver mitochondria is presented. The model represents the system of three ordinary differential equations, and the basic components included are ANT, F_0/F_1 -ATPase, and the phosphate carrier. The model reproduces quantitatively the relationship between mitochondrial membrane potential and the ATP–ADP steady-state exchange rate mediated by the ANT operating in the forward mode, with the assumption that the phosphate carrier functions under rapid equilibrium. Furthermore, the model can simulate the kinetics of experimentally measured data on mitochondrial membrane potential titrated by an uncoupler. Verified predictions imply that the ADP influx rate is highly dependent on the mitochondrial membrane potential, and in the 0–100 mV range it is close to zero, owing to extremely low matrix ATP values. In addition to providing theoretical values of free matrix ATP and ADP, the model explains the diminished ADP–ATP exchange rate in the presence of nigericin, a condition in which there is hyperpolarization of the inner mitochondrial membrane at the expense of the mitochondrial Δ pH gradient; the latter parameter influences matrix inorganic phosphate and ATP concentrations in a manner also described.

Introduction

Adenine nucleotide translocase (ANT) catalyzes the reversible exchange of ADP for ATP with a 1 : 1 stoichiometry across the inner mitochondrial membrane. In this study, we model the ADP–ATP exchange rate during steady state mediated by ANT as a function of mitochondrial membrane potential ($\Delta\Psi_m$). Input data were used from the recently published method exploiting the differential affinity of ADP and ATP for Mg^{2+} [1]. In this method, the rate of appearance of ATP in

the medium following addition of ADP to energized mitochondria is calculated from the measured rate of change in free extramitochondrial $[Mg^{2+}]$ revealed by the membrane-impermeable $5K^+$ salt of the Mg^{2+} -sensitive fluorescent indicator, magnesium green (MgG), using standard binding equations. The assay is designed such that ANT is the sole mediator of changes in $[Mg^{2+}]$ in the extramitochondrial volume as a result of ADP–ATP exchange.

Abbreviations

ANT, adenine nucleotide translocase; A_p5A , diadenosine pentaphosphate; cATR, carboxyatractyloside; MgG, magnesium green; PMF, protonmotive force; SEM, standard error of the mean; $\Delta\Psi_m$, mitochondrial membrane potential.

In this article, we present a kinetic model of mitochondrial phosphorylation, which consists of: (a) the model of adenine nucleotide exchange across the mitochondrial membrane by Metelkin *et al.* [2]; (b) the model of F_0/F_1 -ATPase developed previously by Demin *et al.* [3], (c) the simple steady-state model of the phosphate carrier; and (d) the empirical description of membrane potential formation and ion leakage across the inner mitochondrial membrane. The present model is then validated using data obtained from intact isolated rat liver mitochondria. In addition to providing several predictions elaborated below, this work serves as a complete ATP phosphorylation model that could be incorporated in future versions of larger and more complex models of mitochondrial functions, such as those described recently [4,5].

Results and Discussion

Correlation of ATP–ADP steady-state exchange rate mediated by ANT with $\Delta\Psi_m$

Data for Fig. 1 were obtained from the recently published paper by Chinopoulos *et al.* [1]. Open circle symbols represent the $\Delta\Psi_m$ values reached 20 s after addition of ADP in the presence of increasing concentrations of SF 6847, as detailed in [1]. SF 6847 is a protonophoric uncoupler that dissipates $\Delta\Psi_m$ in a dose-dependent manner, by allowing re-entry of protons into the matrix, bypassing F_0/F_1 -ATP synthase [6]. The dotted line shows the result of the modeling after estimation of the unknown parameters. The con-

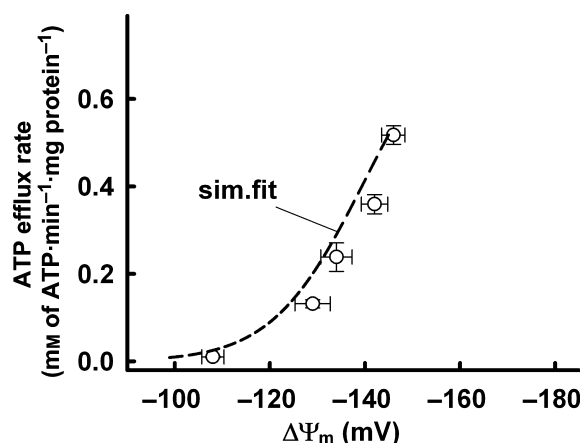


Fig. 1. Correlation of ATP–ADP steady-state exchange rate mediated by ANT with $\Delta\Psi_m$. Plot of ATP–ADP exchange rate mediated by ANT versus $\Delta\Psi_m$ in liver mitochondria depolarized to various voltages by different amounts of SF 6847, constructed from the data as described in [1]. The dashed line represents the result of the model described in the text.

ditions of the described set of experimental data (namely the low concentration of ATP) prevent the reverse functioning of ANT. In that case, the model shows that the synthesis of ATP occurs at a potential from -100 mV or higher. It is important to note that, in this range, mitochondrial ATP production does not saturate; this means that, within the physiological range, ATP production is controlled by $\Delta\Psi_m$. At membrane potential values from 0 mV to -100 mV, the rate of ATP production by mitochondria is close to zero.

Calibration of the kinetic model of phosphorylation in mitochondria

As mentioned in Experimental procedures, there are two parameters of the kinetic model of phosphorylation in mitochondria whose values cannot be estimated on the basis of *in vitro* data obtained with purified enzymes. These are: (a) the value characterized by the activity of ATP synthase (c_{SYN}); and (b) the value characterized by the amount of ANT (c_{ANT}) for a given tissue. These parameters characterize a particular suspension of mitochondria (type of animal, organ, experimental procedure), and require experimental data obtained with this mitochondrial suspension to be identified. To estimate these two parameters, we have fitted our model described above against the dependence of the ATP–ADP steady-state exchange rate mediated by ANT on $\Delta\Psi_m$ (Fig. 1, open circles) and the dependence of these rates on carboxyatractyloside (cATR), a noncompetitive blocker of ANT [7] (Fig. 2, filled circles), measured on a suspension of mitochondria respiring on glutamate and malate. Values of c_{SYN} and c_{ANT} have been chosen (Table 1) in such a way as to provide minimal deviation between experimental data (circles) and the model-generated curve. As a criterion of fitness, the following function was used:

$$f(k_j, K_j) = \sum_i^n \left(\frac{v_i - \tilde{v}_i}{v_i} \right)^2 \quad (1)$$

Here, n is the total number of the experimental points, $\tilde{v}_i$ is the experimentally measured value of the ATP–ADP steady-state exchange rate mediated by ANT, and v_i is the value of the ATP–ADP steady-state exchange rate mediated by ANT calculated on the model at a point corresponding to the experimental one. To estimate values of unknown parameters, the relative error of the model ($\sqrt{f/n}$) has been minimized. This procedure was performed in the DBSOLVE 7 package [8,9] using the Hooke–Jeeves method [10].

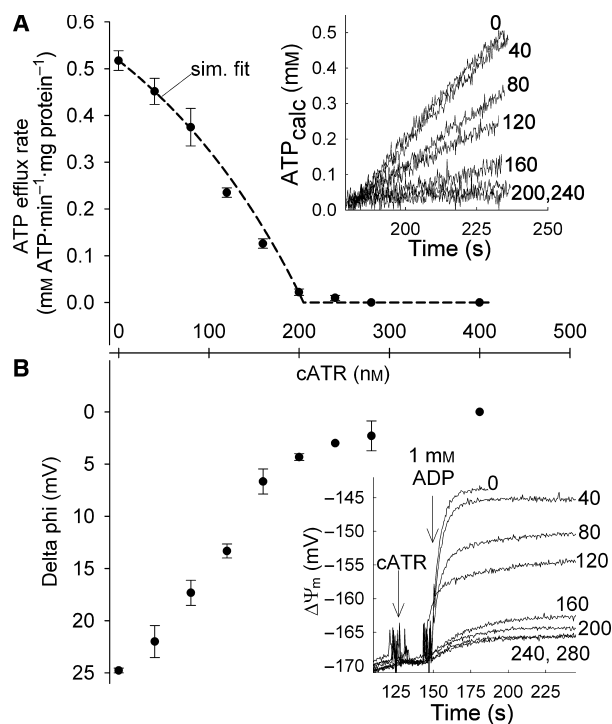


Fig. 2. Titration of ATP–ADP steady-state exchange rate mediated by ANT with cATR and correlation with $\Delta\Psi_m$. (A) ATP–ADP steady-state exchange rate mediated by ANT determined as a function of cATR concentration. Dashed line: simulation fit as described in the text. Inset: a representative experiment showing the calculated [ATP] appearing in the extramitochondrial medium after addition of ADP, in the presence of cATR (in the concentrations indicated in the inset figure, in nM). (B) Delta phi represents the difference in $\Delta\Psi_m$ before and after addition of 1 mM ADP to liver mitochondria pretreated with cATR in the same concentration range as in (A). Inset: a representative experiment showing the effect of the addition of cATR (in the concentrations indicated in the inset of (B), in nM) on $\Delta\Psi_m$, as indicated in the inset of (B). Data in (A) and (B) are shown as SEM from four independent experiments.

Nigericin decreases the ATP–ADP steady-state exchange rate mediated by ANT

Nigericin is an ionophore that mediates the electrically neutral exchange of potassium ions for protons, eliminating the pH gradient across the mitochondrial membrane and causing a compensatory increase in $\Delta\Psi_m$ [11,12]. As seen in Fig. 3, nigericin (10 μ M) decreased the ATP–ADP steady-state exchange rate mediated by ANT significantly, even though it hyperpolarized mitochondria by 15 mV. This is also predicted by the model. We have explained this finding in terms of a decrease in P_i flux through the inner mitochondrial membrane, due to the collapse of Δ pH caused by nigericin. This means that a decrease in $[P_i]$, in turn reducing ATP synthase activity, contributes more to the

steady-state phosphorylation rate than the increase of electric potential and corresponding increase in ATP–ADP steady-state exchange rate mediated by the ANT. As also seen in Fig. 3, the calculated values of the protonmotive force (PMF) in the presence of nigericin are higher than those in the absence of the ionophore. The calibration of the safranin O fluorescence signal may be unreliable in the very highly polarized range, greater than -170 mV [13]; attempts to produce higher membrane potentials (such as by addition of nigericin to fully charged mitochondria) result in deviations from a straight line. This is presumably due to the fact that estimated extramitochondrial K^+ is considered as added K^+ . Thus, $\Delta\Psi_m$ will be overestimated at the point where the concentration of added K^+ approaches that of K^+ that has leaked out from the mitochondria.

Predictions of the kinetic model of phosphorylation in mitochondria: matrix ATP and ADP values and the dependence of P_i on $\Delta\phi$

On the basis of the model developed above and verified against experimental data measured on isolated mitochondria, we have calculated the dependence of matrix concentrations of ADP and ATP as a function of electrical potential difference across the inner mitochondrial membrane. As shown in Fig. 4A, predictions of our model correspond to the experimentally measured (open circles) dependence of O_2 consumption (V_{O_2} in the model) on electric potential difference ($\Delta\Psi_m$). Moreover, our model predicts that concentrations of ADP and ATP (Fig. 4B) at state 3 ($\Delta\Psi_m$ is about -145 mV) are equal to 8.7 mM and 3.3 mM, respectively, and transition from state 3 to state 4 ($\Delta\Psi_m$ is about -170 mV) reverses the order of the concentrations to 2.2 mM for ADP and 9.8 mM for ATP. In order to compare these predicted values with experimental data, we measured matrix ATP and ADP concentrations from mitochondrial matrix extracts by HPLC. Representative traces of HPLC raw data (absorbance at 260 nm versus retention time) are shown in Fig. 4C. AMP, ADP and ATP have been resolved on the basis of different retention times through the HPLC column, identified and calibrated by ‘spiking’ the samples with known amounts of AMP, ADP, and ATP, individually. Assuming 1 μ L of matrix volume for every milligram of mitochondrial protein, we estimated the following values. At 0 mV (no substrates, in the presence of 1 μ M SF 6847), rat liver mitochondria have 3.64 ± 0.34 mM AMP, 8.23 ± 0.65 mM ADP, and 0.51 ± 0.05 mM ATP. At -170 mV (mitochondria energized with 5 mM

Table 1. The parameters of the model.

Parameter	Value	Comment	Source
pH _o	7.25	pH in experimental volume	Measured or given value
pH _i	7.30	pH in matrix under phosphorylating conditions	
[Mg ²⁺] _o ^t	1 mM	Total Mg ²⁺ concentration in experimental volume	
V _o	2 mL	Experimental volume	
[P _i] _o ^t	10 mM	Total inorganic phosphate concentration in experimental volume	
T _o ^t , D _o ^t	Dependent on experimental conditions	Concentration of adenine nucleotides (ATP, ADP) in experimental volume	
[Mg ²⁺] _i	0.35 mM	Buffered Mg ²⁺ concentration in the matrix	[57]
A _i ^t = T _i ^t + D _i ^t	12 mM	Total concentration of adenylates (ATP + ADP) in the matrix (may vary considerably in the range 2.7–22 mM; see [58]).	[58–62]
K _{P,H}	6.31 × 10 ^{−5} mM	Dissociation constant for H ⁺ and phosphate	Calculated from pK _a = 7.2 [63]
K _{T,Mg}	0.114 mM	Dissociation constant for Mg ²⁺ and ATP	[1]
K _{D,Mg}	0.906 mM	Dissociation constant for Mg ²⁺ and ADP	
K _{hyd} ^{SYN}	2.23 × 10 ⁸ mM	Equilibrium constant of ATP hydrolysis	Calculated from ΔG ₀ ' = −30.5 kJ·mol ^{−1} [64]
C _{ANT}	4.8 × 10 ¹ nmol·mg ^{−1}	Effective coefficient (characterizes the amount of ANT dimer per mg of total mitochondrial protein)	Estimated on the basis of fitting of the model against our data
k ₂ ^{ANT,0}	10.8 min ^{−1}	Constant of direct ANT exchange	[2]
k ₃ ^{ANT,0}	21.0 min ^{−1}	Constant of reverse ANT exchange	
K _{T_o} ^{ANT,0}	0.057 mM	Dissociation constant of ATP and ANT	
K _{D_o} ^{ANT,0}	0.051 mM	Dissociation constant of ADP and ANT	
a ₁	0.268	Parameters of ANT electrostatic profile	
a ₂	−0.205		
a ₃	0.187		
δ _T	0.070		
δ _D	0.005		
C _{SYN}	22	Correction factor characterizing activity of ATP synthase in a particular mitochondrial preparation	Estimated on the basis of fitting of the model against our data
n _{SYN}	3	H ⁺ /ATP ratio	[65]
χ	0.9	Parameters of H ⁺ -ATP synthase electrostatic profile	[3]
χ _n	0.1		
V _{max} ^{SYN}	1.2 × 10 ^{−4} nmol (min·mg) ^{−1}	Parameters of H ⁺ -ATP synthase model	
K _{H_o} ^{SYN}	3 × 10 ^{−5} mM		
K _{H_i} ^{SYN}	1 × 10 ^{−6} mM		
K _{MgD} ^{SYN}	5.56 × 10 ^{−3} mM		
K _{P₁} ^{SYN}	3.55 × 10 ^{−1} mM		
K _{MgT} ^{SYN}	9.26 × 10 ^{−1} mM		
F	9.64 × 10 ⁴ C·mol ^{−1}	Faraday constant	
R	8.31 J·mol ^{−1} ·K ^{−1}	Universal gas constant	
T	310 K	Temperature	Measured
C _m	7.8 × 10 ^{−6} F·mg ^{−1}	Capacitance of inner mitochondrial membrane	[66]
k _{O₂}	250 nmol (min·mg) ^{−1}	The empirical coefficients of membrane potential generation	Fitted to experimental data
K _{O₂}	1.45 × 10 ^{−12}		
β _{O₂}	0.36		
k _{leak}	0.438 nmol (min·mg) ^{−1}	The empirical coefficients of membrane leakage description	
β _{leak}	1.05		

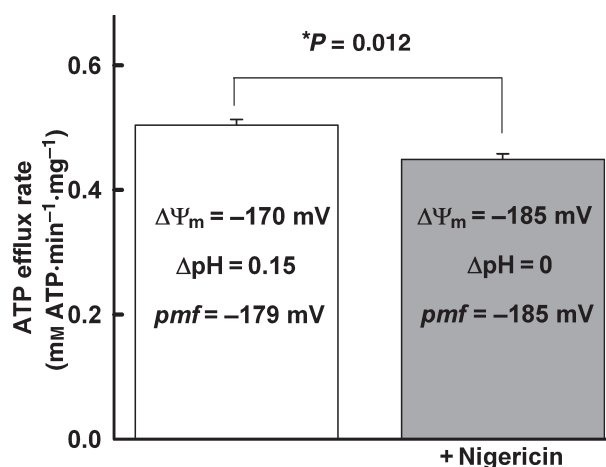


Fig. 3. Effect of nigericin on ATP–ADP steady-state exchange rate mediated by ANT. Bar graph of ATP–ADP steady-state exchange rate mediated by ANT in the absence (white bar) and presence (gray bar) of 10 μM nigericin. PMF shown in the bars was calculated as follows: $PMF = \Delta\Psi_m - 60\Delta pH$ (at 37 °C). Data are shown as SEM from four independent experiments.

glutamate + 5 mM malate), rat liver mitochondria have 2.57 ± 0.67 mM AMP, 2.98 ± 0.41 mM ADP (predicted 2.2 mM), and 7.11 ± 1.55 mM ATP (predicted

9.8 mM). Concerning measuring matrix ATP and ADP values during state 3, this requires addition of ADP to the mitochondrial suspension, followed by conversion of ATP. This creates a technical challenge, because the matrix volume is 2000 times smaller than the experimental volume, and therefore matrix adenylate concentrations are many-fold lower than that in the extramitochondrial compartment. Such obstacles have been addressed by centrifuging mitochondria while phosphorylating through lipid layers, thus excluding as much as possible the water-soluble extramitochondrially located nucleotides, with or without accounting for nucleotides residing in the intermembrane space that would be carried along the lipid layer (e.g. silicon oil). For isolated rat liver mitochondria and using experimental procedures similar – if not identical – to ours, other investigators report a wide range of matrix ATP/ADP ratios during state 3, ranging from 0.01 to 4.5 [14–25] or even in the 8–12 range [26,27]. For mitochondria *in situ* or *in vivo*, most investigators agree with the 1–3 ratio range [28–30]. Those investigators who do not pass isolated mitochondria through silicone oil or do not make corrections for intermembrane space adenine nucleotide retention report matrix ATP/ADP ratios towards the higher values (3–4.5, e.g.

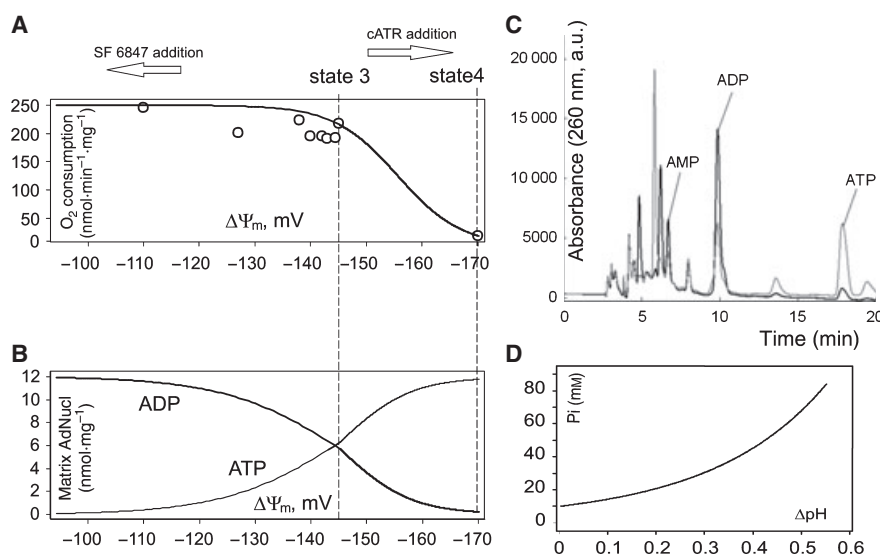


Fig. 4. Steady-state simulations of main characteristics of mitochondria using the model described in the text. (A, B) Experimental conditions described in [1] have been simulated by assigning the following values to model parameters: $[ATP]_{out} = 0$ mM, $[P_i]_{out} = 10$ mM, $pH_{out} = 7.25$, $pH_{in} = 7.35$, $[Mg^{2+}]_{in} = 0.35$ mM, $[Mg^{2+}]_{out}^{total} = 1$ mM. State 3 corresponds to addition of ADP to the experimental volume ($[ADP]_{out} = 1$ mM). State 4 corresponds to addition of cATR at high concentration (full inhibition of ANT). The uncoupling by SF 6847 (left part of the curves) corresponds to an increase in the parameter k_{leak} in the model. (A) Dependence of membrane potential generation rate in terms of O_2 consumption rate. (B) Model-predicted dependence of steady-state concentrations of matrix ADP and ATP on electrical potential differences in the -85 mV to 170 mV $\Delta\Psi_m$ interval. (C) Representative traces of HPLC raw data (from $n = 4$) for the following metabolic conditions: Black line: mitochondria probed without substrates, in the presence of $1 \mu M$ SF 6847. Gray line: mitochondria energized with 5 mM glutamate and 5 mM malate. (D) Model-predicted dependence of matrix phosphate concentration on the difference in pH between the matrix and extramitochondrial space at the following values of model parameters: $[ADP]_{out} = 1$ mM, $[ATP]_{out} = 0$ mM, $[P_i]_{out} = 10$ mM, $pH_{out} = 7.25$.

[31]). Also, it is possible that results obtained after separation of intramitochondrial and extramitochondrial compartments are not relevant, because of the time used for the separation process and possible interconversion of adenine nucleotides even in the presence of inhibitors [22–24,32]. Furthermore, a great proportion of the matrix adenine nucleotides is bound to proteins [33], a notion supported by the fact that rat liver mitochondria retain more than 50% of their total adenine nucleotide content after permeabilization by toluene [34]. Because of this potential binding of adenine nucleotides to intramitochondrial proteins [35–38], the relationship between the measured total ATP/ADP ratio and free intramitochondrial ATP/ADP ratio is difficult to predict. Previous data of Vignais show that a large fraction (75–80%) of the ATP produced by phosphorylation of added ADP within the inner mitochondrial membrane is released into the matrix space before being transported out from the mitochondria; only a small part (20–25%) is released directly outside the mitochondria without penetrating the matrix space [17]. It is therefore inferred that there are separate intramitochondrial pools of adenine nucleotides, one near the ANT and the ATPase, and another located in the bulk of the matrix. The notion of matrix microcompartmentation of adenine nucleotides emanated from several laboratories [17,39–42], but is not accepted unequivocally by several investigators in the field [23,43,44]. Furthermore, microcompartmentation implies the existence of an ATP ‘synthasome’, (ATPase/P_i transporter/ANT in 1 : 1 : 1 ratio), and this is at odds with an estimated ANT/P_i transporter ratio of 4; for a detailed assessment on this matter, the reader is referred to a recent review by Klingenberg [45]. The ability of our model to calculate concentrations of intramitochondrial nucleotides on the basis of $\Delta\Psi_m$ value and values of extramitochondrial ADP and ATP makes it possible to use the model as a toolkit for the study of responses of the intramitochondrial characteristics to external influences [46]. Furthermore, one more prediction that we have derived on the basis of the model is the dependence of the intramitochondrial concentration of P_i on ΔpH . As shown in Fig. 4D, the concentration of matrix P_i can be increased substantially, owing to an increase in ΔpH .

Predictions of the direct-reverse profile of ADP–ATP exchange by ANT as a function of $\Delta\Psi_m$

Mitochondria with nonfunctional respiratory chains become ATP consumers, maintaining an appreciable PMF by pumping protons out of the matrix through the F₀/F₁-ATPase, at the expense of ATP hydrolysis.

Under these conditions, ANT reverses, bringing ATP into the matrix in exchange for ADP, driven by a $\Delta\Psi_m$ less negative than approximately –100 mV [2]. The directionality of ANT is thermodynamically governed by the concentrations of free nucleotides (ATP^{4–} and ADP^{3–}) across the inner mitochondrial membrane, according to Eqn (11).

The concentrations of free ATP^{4–} and ADP^{3–} can be estimated as follows:

$$L = \left(L_t / \left(1 + \frac{Mg^{2+}}{K_{M,app}} \right) \right) / \left(1 + \frac{H^+}{K_H} \right) \quad (2)$$

Here, L denotes ATP^{4–}, L_t denotes the total measured ATP concentration (i.e. ATP^{4–} + ATP-H^{3–} + ATP-Mg^{2–} + ATP-H-Mg[–]), and Mg²⁺ is free magnesium. K_H is the dissociation constant for the reaction ATP-H^{3–} ↔ ATP^{4–} + H⁺, and $K_{M,app}$ is the apparent dissociation constant of MgATP that we have measured at pH 7.25 and $T = 37^\circ\text{C}$. Similarly, the concentration of free ADP^{3–} can also be obtained using Eqn (2), where L is ADP^{3–}, L_t is total ADP concentration (i.e. ADP^{3–} + ADP-H^{2–} + ADP-Mg[–] + ADP-H-Mg), and $K_{M,app}$ is the apparent dissociation constant of MgADP that we have measured at pH 7.25 and $T = 37^\circ\text{C}$. However, the values for K_H and $K_{M,app}$ might be hard to determine for the conditions found inside the matrix. On the basis of the kinetic model, we can estimate the steady-state directionality of ANT on the basis of any given values of [ATP]_o, [ADP]_o, and $\Delta\Psi_m$ (Fig. 5A). With regard to this, it would be useful to construct an experimentally derived $\Delta\Psi_m$ versus ADP–ATP exchange rate profile for the 0–100 mV range; however, it is difficult to establish the relationship of $\Delta\Psi_m$ to ATP consumption rates, because upon exceeding the reversal potential of ANT (indicated by a dotted line in Fig. 5B), $\Delta\Psi_m$ is not clamped at relatively steady states (Fig. 5B, gray curves).

Kinetic behavior of the model resulting from consecutive addition of uncoupler and ADP

$\Delta\Psi_m$ has been shown to fluctuate as a function of time [47–50]; therefore, we sought to formulate our model in order for it to be capable of simulating the time-dependent response of mitochondria to different $\Delta\Psi_m$ values. Titration of $\Delta\Psi_m$ to different values was achieved with different doses of SF 6847 and ADP. To test the applicability of our model for describing the time response, we calculated the time dependencies of electrical potential differences resulting from consecu-

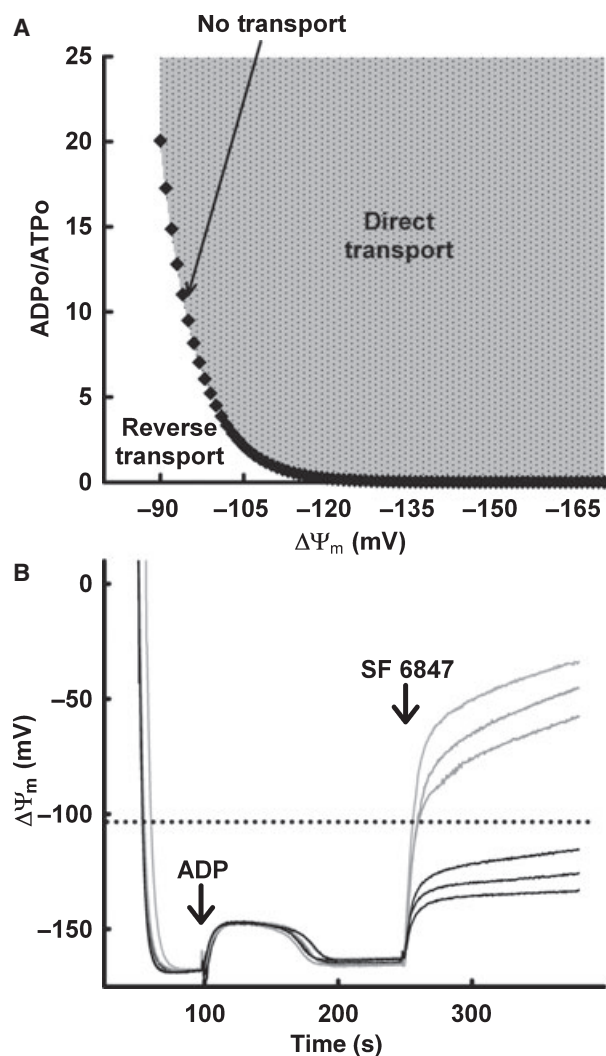


Fig. 5. Forward–reverse profile of ATP/ADP transport and effect of bioenergetic inhibition on $\Delta\Psi_m$. (A) Diagram of directionality of nucleotide transport in mitochondria. Each point of the curve corresponds to the values of $[\text{ADP}]_{\text{out}}/[\text{ATP}]_{\text{out}}$ and $\Delta\Psi_m$ providing 'zero' steady-state flux of adenine nucleotides. The areas above and below the curve correspond to the values of $[\text{ADP}]_{\text{out}}/[\text{ATP}]_{\text{out}}$ and $\Delta\Psi_m$ defining direct and reverse transport of ATP–ADP exchange, respectively. (B) Reconstructed time course of $\Delta\Psi_m$, calculated from safranin O fluorescence. One milligram of liver mitochondria was added to 2 mL of medium and energized by glutamate and malate. ADP (1 mM) was added where indicated, causing a ~ 25 mV depolarization. Upon consumption of ADP, $\Delta\Psi_m$ returns to a level approximating baseline. Increasing concentrations of SF 6847 (10, 20 and 30 nM for the lower three black lines, from bottom to top, and 50, 60 and 70 nM for the upper three gray lines, from bottom to top) were subsequently administered where indicated. The dotted line represents the reversal potential of ANT.

tive addition of uncoupler and ADP to mitochondria in state 2, and compared the results of the calculation with the experimental data presented in [1]. As shown

in Fig. 6A, the model-calculated dependence of $\Delta\Psi_m$ on time corresponds with experimental data (open symbols). The period of time from 0 s to 70 s corresponds to state 2 of mitochondria ($[\text{ADP}]_{\text{out}} = 0$ mM). Different concentrations of uncoupler [Fig. 6A; 30 nM (a), 40 nM (b), 50 nM (c), or 60 nM (d)] were added where indicated. ADP (2 mM) was added after SF 6847, where indicated. As shown in Fig. 6A, the model simulates the steady-state membrane potential sufficiently well, without any fittings of parameters. There is only a slight difference in kinetics upon uncoupler addition at high doses.

To predict the response of mitochondria at state 2 to consecutive addition of uncoupler and ADP, we calculated the time response of ATP efflux (Fig. 6B), O_2 consumption rate (Fig. 6C), and total matrix ADP concentration (Fig. 6D). Time response kinetics of $\Delta\Psi_m$, ATP efflux rate, O_2 consumption rate and ADP in the matrix resulting from the uncoupler or ADP addition depicted in Fig. 6 can be characterized by two features: transition time from one steady state to another, and levels of the steady states. The differences between steady state before and after uncoupler/ADP addition may be characterized by their amplitude. As shown in Fig. 6, the transition time from one steady state to another for all characteristics is less than 10 s.

As shown in Fig. 6A,C,D, the amplitude of time response of electrical potential difference, O_2 consumption rate and total matrix ADP concentration increases with elevation of uncoupler concentration. In contrast, the amplitudes of time responses of $\Delta\Psi_m$ (Fig. 6A), V_{O_2} (Fig. 6C), ATP efflux rate (Fig. 6B) and matrix ADP (Fig. 6D) after ADP (2 mM) addition gradually decrease with increase in the uncoupler concentration.

Experimental procedures

Isolation of mitochondria from rat liver

Mitochondria from rat liver were isolated as detailed previously [51], with minor modifications. All animal procedures were performed according to the guidelines of the local animal care and use committee (Egyetemi Allatkisérleti Bizottság). Briefly, rats were killed, and livers were rapidly removed, chopped, washed extensively, and homogenized using a Teflon–glass homogenizer in ice-cold isolation buffer containing 225 mM mannitol, 75 mM sucrose, 5 mM Hepes, 1 mM EGTA, and 1 mg·mL⁻¹ BSA (fatty acid-free), with the pH adjusted to 7.4 with Tris. The homogenates were centrifuged at 1250 *g* for 3 min; the pellet was discarded, and the supernatant was centrifuged at 12 000 *g* for 10 min; this step was repeated once. At the end of the

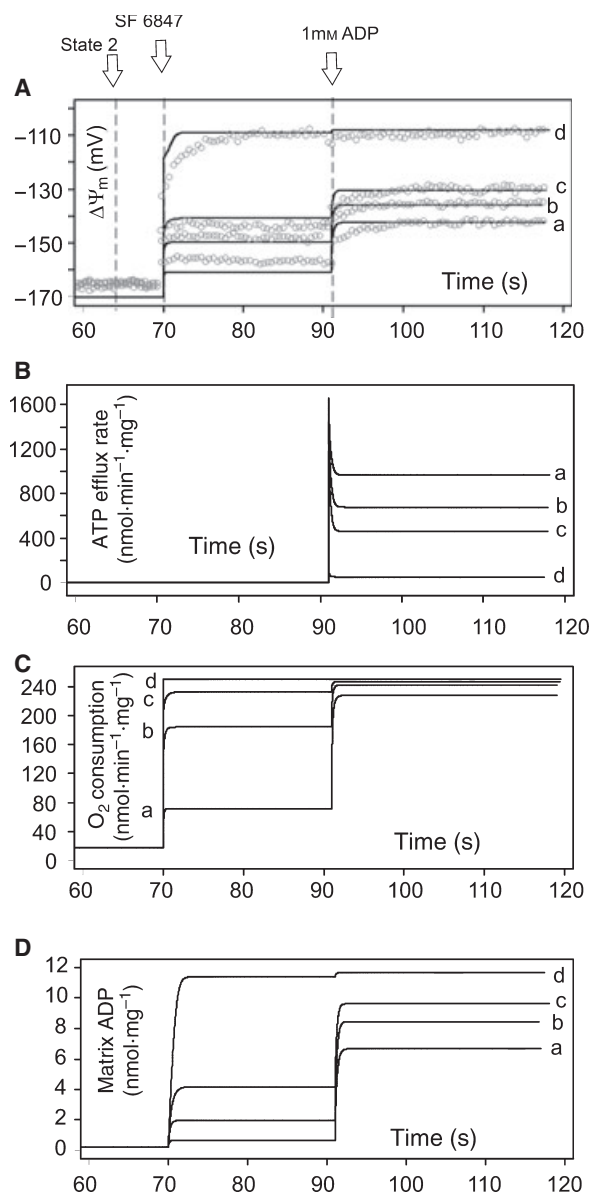


Fig. 6. The kinetics of the main characteristics of mitochondria. (A) Plot of electrical membrane potential versus time. Solid black lines indicate the kinetics of mitochondrial membrane potential. Open symbols indicate the calibrated $\Delta\Psi_m$ data obtained from panel 6C of [1]. (B) Plot of ATP efflux rate versus time. (C) Time dependence of O₂ consumption rate. (D) The kinetics of total matrix ADP concentration. The model parameters have been chosen in such a way as to simulate the experimental data presented in Fig. 1 for different doses of uncoupler. The experimental conditions described in [1] have been simulated by assigning the following values to model parameters: $[\text{ATP}]_{\text{out}} = 0$ mM, $[\text{P}_i]_{\text{out}} = 10$ mM, $\text{pH}_{\text{out}} = 7.25$, $\text{pH}_{\text{in}} = 7.35$, $[\text{Mg}^{2+}]_{\text{in}} = 0.35$ mM, $[\text{Mg}^{2+}]_{\text{out}}^{\text{total}} = 1$ mM. The initial period (0–70 s) describes a steady state corresponding to state 2 of mitochondria ($[\text{ADP}]_{\text{out}} = 0$ mM). After 70 s, different doses of the uncoupler SP 6847 were added. At time 90 s, the ADP was added to the experimental volume. Letters a, b, c, d correspond to different uncoupler doses: 30 nM (a), 40 nM (b), 50 nM (c), or 60 nM (d).

second centrifugation, the supernatant was discarded, and the pellet was suspended in 500 μL of the same buffer without EGTA. The mitochondrial protein concentration was determined using the Biuret assay.

$[\text{Mg}^{2+}]_f$ determination from MgG fluorescence in the extramitochondrial volume of isolated mitochondria and conversion to ADP–ATP exchange rate

Mitochondria (1 mg, wet weight; in this and all subsequent experiments, wet weight of mitochondrial amount is implied) were added to 2 mL of an incubation medium containing 8 mM KCl, 110 mM potassium gluconate, 10 mM NaCl, 10 mM Hepes, 10 mM KH_2PO_4 , 0.005 mM EGTA, 10 mM mannitol, 1 mM MgCl_2 , 5 mM glutamate, 5 mM malate, 0.5 $\text{mg}\cdot\text{mL}^{-1}$ BSA (fatty acid-free) (pH 7.25), 50 μM diadenosine pentaphosphate (A_p5A), and 2 μM MgG 5K^+ salt. Including the adenylate kinase inhibitor A_p5A in the medium is essential; Mg^{2+} , which is present in the assay medium, activates adenylate kinase. A_p5A is a potent inhibitor of adenylate kinase [52]. MgG fluorescence was recorded in a PTI Deltascan fluorescence spectrophotometer at a 5 Hz acquisition rate, using 506 nm and 530 nm excitation and emission wavelengths, respectively. MgG exhibits an extremely high quantum yield ($E_M[\text{MgG}] = 75\,000\text{--}100\,000\text{--}1\cdot\text{cm}^{-1}$); therefore, slits were opened to widths of no more than 1 nm. Experiments were performed at 37 °C. At the end of each experiment, minimum fluorescence (F_{min}) was measured after addition of 4 mM EDTA, and this was followed by the recording of maximum fluorescence (F_{max}) elicited by addition of 20 mM MgCl_2 . Free Mg^{2+} concentration ($[\text{Mg}^{2+}]_f$) was calculated from the equation $[\text{Mg}^{2+}]_f = [K_d(F - F_{\text{min}})/(F_{\text{max}} - F)] - 0.055$ mM, assuming a K_d of 0.9 mM for the $\text{MgG}\text{--}\text{Mg}^{2+}$ complex [53]. The correction term -0.055 mM is empirical, and possibly reflects chelation by EDTA of other ions that have an affinity for MgG, and alter its fluorescence. This term was needed to obtain a reliable $[\text{Mg}^{2+}]$ estimate, as determined from calibration experiments using solutions with known, stepwise increasing, Mg^{2+} concentrations. ADP–ATP exchange rate was estimated using the recently described method by our laboratory [1], exploiting the differential affinity of ADP and ATP for Mg^{2+} . The rate of ATP appearing in the medium following addition of ADP to energized mitochondria (or vice versa in the case of de-energized mitochondria) is calculated from the measured rate of change in free extramitochondrial $[\text{Mg}^{2+}]$ using standard binding equations. The assay is designed such that ANT is the sole mediator of changes in $[\text{Mg}^{2+}]$ in the extramitochondrial volume, as a result of ADP–ATP exchange [1]. For the calculation of $[\text{ATP}]$ or $[\text{ADP}]$ from free $[\text{Mg}^{2+}]$, the apparent K_d values are identical to those in [1], owing to identical experimental conditions ($K_{\text{ADP}} = 0.906 \pm 0.023$ mM, and $K_{\text{ATP}} = 0.114 \pm 0.005$ mM).

$\Delta\Psi_m$ determination in isolated mitochondria

$\Delta\Psi_m$ was estimated using fluorescence quenching of the cationic dye safranin O due to its accumulation inside energized mitochondria [13]. Mitochondria (1 mg) were added to 2 mL of an incubation medium containing 8 mM KCl, 110 mM potassium gluconate, 10 mM NaCl, 10 mM Hepes, 10 mM KH_2PO_4 , 0.005 mM EGTA, 10 mM mannitol, 1 mM MgCl_2 , 5 mM glutamate, 5 mM malate, 0.5 $\text{mg}\cdot\text{mL}^{-1}$ BSA (fatty acid-free) (pH 7.25), 50 μM A_p5A , and 5 μM safranin O. All of the experiments were performed in the presence of cyclosporin A. Parallel experiments in the absence of this compound verified that it did not interfere with the outcome. Fluorescence was recorded in a Hitachi F-4500 spectrofluorimeter (Hitachi High Technologies, Maidenhead, UK) at a 2 Hz acquisition rate, using 495 nm and 585 nm excitation and emission wavelengths, respectively. Experiments were performed at 37 °C. To convert safranin O fluorescence into millivolts, a voltage–fluorescence calibration curve was constructed. To this end, safranin O fluorescence was recorded in the presence of 2 nM valinomycin and with stepwise increases in $[\text{K}^+]$ (in the 0.2–120 mM range), which allowed calculation of $\Delta\Psi_m$ by the Nernst equation, assuming a matrix $[\text{K}^+]$ of 120 mM [13].

Mitochondrial oxygen consumption

Mitochondrial respiration was recorded at 37 °C with a Clark-type oxygen electrode (Hansatech, King's Lynn, UK). Mitochondria (1 mg) were added to 2 mL of an incubation medium containing 8 mM KCl, 110 mM potassium gluconate, 10 mM NaCl, 10 mM Hepes, 10 mM KH_2PO_4 , 0.005 mM EGTA, 10 mM mannitol, 1 mM MgCl_2 , 5 mM glutamate, 5 mM malate, 0.5 $\text{mg}\cdot\text{mL}^{-1}$ BSA (fatty acid-free) (pH 7.25), and 50 μM A_p5A . State 3 respiration was initiated by the addition of 1 mM K-ADP to the incubation medium. State 4 respiration was initiated by the addition of cATR at the indicated concentrations.

Determination of matrix adenine nucleotides by HPLC

Rat liver mitochondria (0.25 mL of 65 $\text{mg}\cdot\text{mL}^{-1}$) were added to 1 mL of buffer with 5 mM glutamate and 5 mM malate or without substrates (but in the presence of 1 μM SF 6847) containing 8 mM KCl, 110 mM potassium gluconate, 10 mM NaCl, 10 mM Hepes, 10 mM KH_2PO_4 , 0.005 mM EGTA, 10 mM mannitol, 1 mM MgCl_2 , 0.5 $\text{mg}\cdot\text{mL}^{-1}$ BSA (fatty acid-free) (pH 7.25), 1 μM cyclosporin A (to inhibit opening of the permeability transition pore, which could lead to loss of matrix adenylate nucleotide pools [54]) and 50 μM A_p5A for 3 min. Subsequently, 1 mL of this mixture was added to 1 mL of ice-cold perchloric acid (3 M), and allowed to deproteinize at 0 °C for

5 min. After this, 2 mL of 1.5 M KOH and 0.5 M Tris was added, and the precipitate was allowed to form at 0 °C for another 5 min. Then, 0.8 mL of the supernatant was spun at 25 000 *g* for 3 min at 4 °C, and 0.6 mL was collected, adjusted to pH 6.5–6.7 with perchloric acid or KOH and Tris, and respun at 25 000 *g* for 3 min at 4 °C to remove any remaining precipitate. Supernatants were immediately frozen with liquid nitrogen, and were kept at –70 °C for further use. The chromatographic separation of adenine nucleotides (AMP, ADP, and ATP) was performed with a C18 reversed-phase column (ODS Hypersyl; 250 × 4.6 mm internal diameter; particle size 5 μm). The mobile phase was composed of 215 mM sodium dihydrogen phosphate, 2.3 mM tetrabutylammonium hydroxide, 4% acetonitrile, and 0.4% potassium hydroxide, and the flow rate was 1 $\text{mL}\cdot\text{min}^{-1}$. The sample injection volume was 20 μL , and during isocratic acquisition the components were monitored at 260 nm with a multiwavelength Jasco Pu-2075 Plus Intelligent UV detector connected to a Jasco Pu-2089 Quaternary Gradient pump and Rheodyne sample injector (Jasco, Gross-Umstadt, Germany). Calibration of the signals was performed by 'spiking' the samples with known amounts of AMP, ADP and ATP in a relevant range of concentrations.

Kinetic model of phosphorylation in mitochondria

The kinetic model of the phosphorylation subsystem of mitochondrial oxidative phosphorylation includes a quantitative description of the following processes: (a) ATP synthesis, catalyzed by ATPase/ATP synthase (V_{SYN}); (b) electrogenic translocation of adenine nucleotides, catalyzed by the adenine nucleotide translocase (V_{ANT}); (c) electro-neutral symport of P_i and a proton, as catalyzed by the phosphate carrier; (d) electrogenic transport of protons from the matrix to the intermembrane space by the electron transport chain (complexes I–IV) with generation of membrane potential (V_{O_2}); and (e) leakage of K^+ , H^+ and other ions across the mitochondrial inner membrane (V_{leak}).

The transport and synthesis of adenylates can be represented by system of algebra-differential equations:

$$\begin{cases} \frac{d}{dt} D_i^t = -V_{\text{SYN}} + V_{\text{ANT}}, \\ T_i^t + D_i^t = A_i^t. \end{cases} \quad (3a)$$

Here, T_i^t and D_i^t stand for concentrations of total ATP and ADP in the mitochondrial matrix. Additionally, it is necessary to take into account the ionic balance in the system. The total ionic current can be represented as follows:

$$I = F \cdot (20 \cdot V_{\text{O}_2} - V_{\text{leak}} - 3 \cdot V_{\text{SYN}} - V_{\text{ANT}})$$

where I stands for total current (positive and negative) of ions transported across the inner membrane of mitochondria.

The positive current direction is from the matrix to the intra-membrane space. The integer coefficients of the right-hand side of the second equation of the system (3) were chosen on the basis of the knowledge of number of charges transferred or leaked across the inner mitochondrial membrane: (a) the electron transport chain transports 20 protons per O_2 ; (b) F_0/F_1 -ATPase transports three protons per one molecule of ATP synthesized; and (c) one cycle of transport by ANT leads to transport of one additional charge.

The ionic current across the membrane determines the changes in membrane potential.

$$I = C_m \frac{d\Delta\Psi_m}{dt}$$

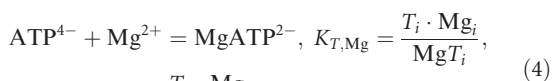
Thus, the changes in membrane potential can be described as follows:

$$\frac{d\Delta\Psi_m}{dt} = \frac{F}{C_m} (20 \cdot V_{O_2} - V_{leak} - 3 \cdot V_{SYN} - V_{ANT}) \quad (3b)$$

Eqn (3a,b) represent the full system describing the phosphorylation of mitochondria.

The following ion balances have been taken into account in the model.

The first is the binding/dissociation of magnesium ions to/from adenylate nucleotides in the mitochondrial matrix and outside of mitochondria to form the complexes $MgADP^-$ and $MgATP^{2-}$:



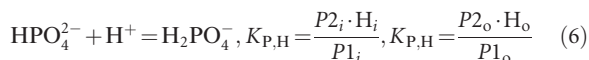
$$K_{T,Mg} = \frac{T_o \cdot Mg_o}{MgT_o}$$



$$K_{D,Mg} = \frac{D_o \cdot Mg_o}{MgD_o}$$

Here, T_o and D_o are the concentrations of free ATP and ADP outside of mitochondria. Values of dissociation constants are listed in Table 1.

The second is the binding/dissociation of protons to/from P_i to form the complexes $H_2PO_4^-$ and HPO_4^{2-} in the mitochondrial matrix and outside of mitochondria.



Here, P_{1i} , P_{2i} and P_{1o} , P_{2o} are the concentrations of twice-protonated and once-protonated P_i in the mitochondrial matrix and outside the mitochondria, respectively. All of these binding/dissociation processes are assumed to be at equilibrium.

On the basis of Eqns (4,5), we can express the concentrations of free adenine nucleotides and their complexes with

magnesium in terms of total concentrations of ADP (D_o^t , D_i^t) and ATP (T_o^t , T_i^t):

$$\begin{aligned} T_i &= T_i^t \frac{1}{1 + \frac{Mg_i}{K_{T,Mg}}}, T_o = T_o^t \frac{1}{1 + \frac{Mg_o}{K_{T,Mg}}} \\ MgT_i &= T_i^t \frac{\frac{Mg_i}{K_{T,Mg}}}{1 + \frac{Mg_i}{K_{T,Mg}}}, MgT_o = T_o^t \frac{\frac{Mg_o}{K_{T,Mg}}}{1 + \frac{Mg_o}{K_{T,Mg}}} \\ D_i &= D_i^t \frac{1}{1 + \frac{Mg_i}{K_{D,Mg}}}, D_o = D_o^t \frac{1}{1 + \frac{Mg_o}{K_{D,Mg}}} \\ MgD_i &= D_i^t \frac{\frac{Mg_i}{K_{D,Mg}}}{1 + \frac{Mg_i}{K_{D,Mg}}}, MgD_o = D_o^t \frac{\frac{Mg_o}{K_{D,Mg}}}{1 + \frac{Mg_o}{K_{D,Mg}}} \end{aligned} \quad (7)$$

Using Eqn (6), we can express the concentrations of extramitochondrial $H_2PO_4^-$ and HPO_4^{2-} in terms of total concentration of P_i [P_i] t

$$\begin{aligned} P_{2o} &= P_o^t \frac{1}{1 + \frac{H_o}{K_{P,H}}} \\ P_{1o} &= P_o^t \frac{\frac{H_o}{K_{P,H}}}{1 + \frac{H_o}{K_{P,H}}} \end{aligned} \quad (8)$$

The P_i/H carrier of mitochondria catalyzes the electro-neutral symport of twice-protonated phosphate and proton: $H_2PO_4^- + H_o^+ = (H_2PO_4^-)_i + H_i^+$. The V_{max} of P_i transport is much higher than the rates of adenylate transport and synthesis [55], and the K_m is much lower than the concentration of P_i inside or outside of the matrix. Thus, the phosphate transport does not limit oxidative phosphorylation under physiological conditions. According to the rapid equilibrium approximation, we can express the concentrations of matrix $H_2PO_4^-$ and HPO_4^{2-} in terms of extramitochondrial phosphate concentration and pH values in the mitochondrial matrix and extramitochondrial space:

$$K_{eq}^{P/H} = \frac{P_{1o} \cdot H_o}{P_{1i} \cdot H_i}$$

So, taking into account the 1 : 1 stoichiometry and electroneutrality of P_i/H transport, we can conclude that $K_{eq} = 1$, so we can write the following:

$$\begin{aligned} P_{1i} &= P_{1o} \cdot \frac{H_o}{H_i} \\ P_{2i} &= P_{1i} \frac{K_{P,H}}{H_i} \end{aligned} \quad (9)$$

Here $[H^+]_i$ and $[H^+]_o$ are the proton concentrations in the mitochondrial matrix and outside of mitochondria:

$$H_o = 10^{3-pH_o} \text{ mM}$$

$$H_i = 10^{3-pH_i} \text{ mM}$$

The rate equation of the ATP synthase reaction is based on a minimal kinetic scheme for ATP synthesis–hydrolysis [3]:

$$V_{\text{SYN}} = c_{\text{SYN}} \cdot V_{\text{max}}^{\text{SYN}} \exp(n_{\text{SYN}} \chi \phi) \left(\frac{H_o}{K_{H_o}^{\text{SYN}}} \right)^{n_{\text{SYN}}} \times \\ \times \frac{1}{K_{MgD}^{\text{SYN}} \cdot K_{P_i}^{\text{SYN}} \left(1 + \frac{MgD_i \cdot P_i}{K_{MgD}^{\text{SYN}} \cdot K_{P_i}^{\text{SYN}}} \left(\frac{H_o}{K_{H_o}^{\text{SYN}}} \right)^{n_{\text{SYN}}} + \frac{MgT_i}{K_{MgT}^{\text{SYN}}} \left(\frac{H_i}{K_{H_i}^{\text{SYN}} \exp(\chi_n \phi)} \right)^{n_{\text{SYN}}} \right)}$$

Here,

$$\phi = -\frac{F\Delta\psi}{RT} \text{ and } K_{\text{eq}}^{\text{SYN}} = K_{\text{hyd}}^{\text{SYN}} \frac{K_{T,Mg}}{K_{D,Mg}} \cdot \frac{10^{-7+3}}{10^{-7+3} + K_{P,H}}$$

Values of all parameters of the equation except c_{SYN} are taken from [3,56] and listed in Table 1 and references therein. c_{SYN} is a dimensionless correction factor characterizing the activity of ATP synthase in the particular mitochondrial preparation. This factor has been estimated through the fitting of the model to the experimental data presented in this article (see Results and discussion).

The rate equation for adenine nucleotide translocation has been derived in [2]:

$$v_{\text{ANT}} = c_{\text{ANT}} \cdot \frac{1}{\Delta^{\text{ANT}}} \left(k_2^{\text{ANT}} q^{\text{ANT}} \frac{T_i \cdot D_o}{K_{D_o}^{\text{ANT}}} - k_3^{\text{ANT}} \frac{D_i \cdot T_i}{K_{T_o}^{\text{ANT}}} \right), \\ \Delta^{\text{ANT}} = \left(1 + \frac{T_o}{K_{T_o}^{\text{ANT}}} + \frac{D_o}{K_{D_o}^{\text{ANT}}} \right) (D_i + q^{\text{ANT}} \cdot T_i),$$

Here,

$$q^{\text{ANT}} = \frac{k_3^{\text{ANT}} K_{D_o}^{\text{ANT}}}{k_2^{\text{ANT}} K_{T_o}^{\text{ANT}}} \exp(\phi), \\ K_{D_o}^{\text{ANT}} = K_{D_o}^{\text{ANT},0} \exp(3\delta_D \phi), \\ K_{T_o}^{\text{ANT}} = K_{T_o}^{\text{ANT},0} \exp(4\delta_T \phi), \\ k_2^{\text{ANT}} = k_2^{\text{ANT},0} \exp\{(-3a_1 - 4a_2 + a_3)\phi\}, \\ k_3^{\text{ANT}} = k_3^{\text{ANT},0} \exp\{(-4a_1 - 3a_2 + a_3)\phi\}$$

Values of all parameters of the equation for except c_{ANT} are taken from [2] and listed in Table 1 and references therein. The value of c_{ANT} refers to the apparent (not the true) concentration of ANT dimers per mg of total mitochondrial protein. Indeed, values for $k_2^{\text{ANT},0}$ and $k_3^{\text{ANT},0}$ have been estimated (see [2] for details) on the basis of the experimental data obtained with proteoliposomes. This means that the values of the rate constants are underestimated, because a proportion of the ANT proteins may be

damaged in that experiment. Consequently, to use the equation of ANT activity, it is necessary to estimate c_{ANT} for the given mitochondrial suspension. In this way, the values of

$$k_{\text{ANT}+}^0 = k_2^{\text{ANT},0} \cdot c_{\text{ANT}}$$

and

$$k_{\text{ANT}-}^0 = k_3^{\text{ANT},0} \cdot c_{\text{ANT}}$$

can be considered as ‘true’ direct and reversed activities of ANT at zero membrane potential. These values characterize the given suspension of mitochondria, but not the value of c_{ANT} itself. Thus, c_{ANT} has been estimated through the fitting of the model to the experimental data presented in this article (see Results and discussion).

In order to describe the generation/consumption of membrane potential, it is necessary to take into account a variety of processes of ion exchange and leaks catalyzed by different enzymes and transporters. Several models have been published for this purpose (e.g. Demin *et al.* [3]). However, to simulate ATP synthesis only in respiration states 2, 3 and 4, a detailed description is obviously redundant. In our study, a simple empirical description of membrane potential of generation/consumption was used. Indeed, we assumed that oxygen consumption rate (V_{O_2}) and the rate of ion leaks through the inner mitochondrial membrane were given by the following equations:

$$V_{\text{O}_2} = \frac{k_{\text{O}_2}}{1 + K_{\text{O}_2} \exp(\beta_{\text{O}_2} \phi)} \\ V_{\text{leak}} = k_{\text{leak}} \exp(\beta_{\text{leak}} \phi)$$

There are five parameters in these equations. The values of the parameters (Table 1) were chosen in such a way as to fit the experimentally measured dependence of O_2 consumption on electrical potential difference depicted in Fig. 4A, and to allow the model to describe values of respiratory rate at states 2, 3 and 4. Indeed, our model was verified against the following experimental data: state 3 corresponds to $V_{\text{O}_2} = 216 \text{ nmol (min}\cdot\text{mg)}^{-1}$, $\Delta\Psi_m = -145 \text{ mV}$; state 2 corresponds to $V_{\text{O}_2} = 19 \text{ nmol (min}\cdot\text{mg)}^{-1}$, $\Delta\Psi_m = -170 \text{ mV}$; and state 4 (induced by cATR) corresponds to $V_{\text{O}_2} = 17 \text{ nmol (min}\cdot\text{mg)}^{-1}$, $\Delta\Psi_m = -170 \text{ mV}$.

Reagents

Standard laboratory chemicals, tetrabutylammonium hydroxide and HPLC-grade acetonitrile were from Sigma (St Louis, MO, USA). A_p5A , safranin O and valinomycin were from Sigma. SF 6847 was from Biomol (catalog number EI-215; BIOMOL GmbH, Hamburg, Germany). All mitochondrial substrate stock solutions were dissolved in double-distilled water and titrated to pH 7.0 with KOH.

ATP, ADP and AMP were purchased as potassium salts of the highest purity available, and titrated to pH 6.9 with KOH.

Statistics

Data are presented as mean $\pm$ standard error of the mean (SEM); significant differences between groups of data were evaluated by one-way ANOVA followed by Tukey's *post hoc* analysis, with $P < 0.05$ being considered significant.

Acknowledgements

We thank Professor Noszál Béla for providing access to the HPLC setup, and Dr Judit Doczi for helpful discussions. This work was supported by the Russian Foundation for Basic Research (grant no. 09-01-12097), the FP7 ETHERPATHS project, to O. Demin, and Semmelweis University Research Grant 63320, OTKA-NKTH Grant NF68294 and OTKA Grant NNF78905 to C.C.

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